TANVIR AHMED

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I am a final-year Computer Science and Engineering student at North South University, working on multimodal machine learning: which inputs actually need an expensive model, and when a model should say that it does not know. One accepted paper, six more under review, and three months as a research assistant at the NSU Trimodal AI Lab. Alongside the research I build full-stack systems, and I have spent time on quantum machine learning and on inference for microcontrollers.

Research Interests

- Efficient multimodal inference
- Computer vision and video understanding
- Calibration, abstention and uncertainty
- Agentic systems
- Quantum machine learning
- Explainable AI
- Edge and embedded machine learning
- Full-stack system architecture

Education

Bachelor of Science in Computer Science and Engineering **Aug 2022 – Dec 2026**

North South University, Dhaka, Bangladesh. Department of Electrical and Computer Engineering. Expected graduation December 2026.
CGPA: **2.99 / 4.00**

Higher Secondary Certificate (HSC), Science **Sep 2019 – Dec 2021**

Milestone College, Dhaka. GPA: **5.00 / 5.00**

Experience

Research Assistant, Trimodal AI Lab **Mar 2026 – May 2026**

North South University, Dhaka, Bangladesh. Contract, on-site, academic research.

- Worked on multimodal and agentic AI systems under faculty supervision: reading the literature, testing ideas and building prototypes.
- Ran experiments with other lab members and wrote up the results, including what they did not show.
- Part of the lab since 2025, through its reading group and joint projects.

Publications

Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency — **Accepted**

Tanvir Ahmed. *LM-KBC 2026 shared task (5th edition), Automated Knowledge Base Construction (AKBC) at EMNLP 2026.*

- Set-valued knowledge base construction with an open-weight model of at most 32B parameters, no retrieval and no fine-tuning. Deciding *when to answer* is treated as a first-class decision, and the system beats the official baseline on five of the six relations.

Code: github.com/minhaz-42/lm-kbc-2026

SoundSentry: A Low-Cost Edge-AI Acoustic Event Detection and Direction-Finding System on STM32 — **Under review**

Tanvir Ahmed, Sanzana Afrin Tonny, Husain Kabir Sifat. *International Conference on Electrical and Computer Engineering (ICECE) 2026, Dhaka.*

- STM32F103 prototype with a three-microphone array, ADC-DMA acquisition, on-device classification over 11 acoustic features, servo-based direction response and SD-card logging.

Code: github.com/minhaz-42/SoundSentry

SimCert: A Classical-Simulability Audit of Trained QNLP Text Models — **Under review**

Submitted to QTML 2026.

- Re-simulates trained quantum-NLP circuits under matrix-product-state truncation, and ablates entanglement, to test whether the quantum component is load-bearing.

Technologies: Python, PennyLane, lambeq, OpenQASM, Hydra

Code: github.com/minhaz-42/simcert-qnlp

Observing Is Not Free: The Decision Value of Visual Observations in Vision-Language Agents — **Under review**

- Asks whether an agent's observation-selection criterion ranks observations the way their realised effect on the answer does, measured with a counterfactual table of the model's answer before and after every candidate observation. Decision value turns out not to be predictable from crop geometry.

Technologies: Python, Vision-Language Models

Knowing What You No Longer Know: Calibrated Belief Validity for Persistent World Memory — **Under review**

- Separates whether a belief is still true from how confident the model was when it formed it, and builds a benchmark of elapsed-time-matched pairs over real and simulated changing rooms to score the difference.

Technologies: Python, PyTorch, HD-EPIC, AI2-THOR, ProcTHOR

Sequential Video Belief — **Under review**

- Tests whether a video-language model's own stated answer impairs its ability to revise that answer when later visual evidence contradicts it, with the go or no-go rule fixed before analysis.

Technologies: Python, Vision-Language Models

Uncertainty-Guided Language-Grounded 3D Scene Graphs from Sparse Views **Under review**

- Frozen geometry and vision-language models with a lightweight graph reasoning module, for object-level 3D segmentation and spatial relations from only a few casual images.

Technologies: Python, PyTorch, VGGT, SAM 2, CLIP, DINOv2

Research Projects

Frame-Aware Multi-Model Scheduling for Efficient Video Captioning **Senior Design, 2026**

Routes video captioning between a 4B and a 27B vision-language model so the large model is spent only where the footage is hard. Lead author of four.

- Frames are stitched into mosaics, encoded by a frozen CLIP ViT-B/32 and scored for visual complexity by a small graph attention network trained without human labels.
- Answers a 19-clip benchmark in fewer requests than either model alone, while scoring above the small model on every judged axis.

Technologies: Python, PyTorch, PyTorch Geometric, CLIP, Django, Gemma 3, Qwen3-VL

Quantum-Enhanced Driver Distraction Detection **Directed Research, 2026**

A hybrid classical–quantum pipeline with uncertainty-aware risk scoring, tested under a protocol where only the classification head differs.

- The variational quantum head reaches accuracy parity with its matched classical control and cuts expected calibration error by 37% relative.
- The report states what the evidence does not support, including that the calibration gain cannot be attributed to the circuit.

Technologies: Python, PyTorch, PennyLane, EfficientNet, scikit-learn

Bangla Mental Health Conversational Agent **Ongoing**

Mental health chatbot for Bangla-speaking users, built on fine-tuned Gemma-3-Bangla and Qwen-2.5-0.5B-Bangla models, with voice input and output, mood tracking, conversation history and a multi-language interface.

Technologies: Python, Django, PyTorch, Transformers, OAuth, Apple MPS, SQLite

Technical Projects

Shadora: Cooking Companion 2026

Tell it what is in your kitchen and it finds what you can cook now, what needs one or two more things, and which single purchase unlocks the most, by exact arithmetic rather than a match percentage. 505 original recipes across 22 cuisines, each scaling exactly and drawn step by step; cooking mode with timers that run across pages; allergens as hard rules; nutrition drawn only from published composition tables, with every value's source kept. Ships as a web app and an Android build, in English and Bangla.

Technologies: Next.js 16, React 19, TypeScript, Tailwind 4, PostgreSQL, Drizzle, Zod, Capacitor, Ollama, Vitest, Playwright

Links: kitchensketch.vercel.app

Dishari: Bank and Card Guide for Bangladesh 2026

Bilingual English and Bangla guide to bank accounts, credit cards and card offers in Bangladesh. Every fee, rate and rule carries the issuer's own document, page, exact wording and the dates it applies. Compares two to four products on their differences, runs a deterministic five-step matcher that explains fit and trade-offs, and computes real yearly costs; an admin data desk reviews changes as old-to-new diffs with an audit trail.

Technologies: Next.js 16, React 19, TypeScript, Tailwind 4, PostgreSQL, Drizzle, Zod, Vitest, Playwright with axe

Pakna Council: AI Council Platform 2026

Local-first web app that runs several LLMs in parallel on one prompt through Ollama, with model comparison and vision workflows.

Technologies: Python, Django, React, Docker, Ollama, WebSocket

Links: Source · Live site

Nestledown: 3D Exploration Game 2026

Cozy 3D exploration game built solo in Godot 4, with every model, texture, animation and sound generated by a script and no third-party asset in the repository. A parallel build of the same world runs on a custom C++20 engine.

Technologies: Godot 4, Blender, C++20, SDL3, CMake

Links: Announcement trailer

SHAND: Explainable Assumption Detector 2026

Rule-based system for finding assumptions in English text, with visual analytics and export.

Technologies: Python, NLP, HTML, JavaScript, D3.js

Links: Source · Live site

ISKA: Privacy-First Document Analysis Assistant 2026

Document analysis assistant with a Chrome extension for use while reading.

Technologies: TypeScript, Chrome extension APIs

Links: Source · Live site

SUMBA: 3D Sign Language Translation 2026

Real-time sign-to-text translation using computer vision, with mobile integration and offline use.

Technologies: Django, MediaPipe, Transformers, React Native

Links: Source

Entangled Fusion Quantum Circuits for Vision–Language Alignment 2026

Experimental multimodal platform that combines vision analysis with quantum-inspired fusion and natural language reasoning.

Technologies: Python, Django, PennyLane, Ollama, OpenCV, REST APIs

Links: Source

FitWell: AI Nutrition Assistant 2025

AI-powered meal planning, BMI tracking and health insights.

Technologies: Django, Python, Qwen, SQLite, Chart.js

Links: Source

DG Healthcare Management System 2024

Hospital management platform with appointments, billing and role-based access.

Technologies: HTML5, CSS3, PHP, MySQL, JavaScript

Links: Source

CSE215 Java Application 2024

Object-oriented Java application with a modular MVC structure and unit tests.

Technologies: Java, JavaFX, MySQL, JUnit

Links: Source

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C/C++, PHP, SQL, HTML, CSS, LaTeX

Machine Learning and AI: Deep learning, computer vision, NLP, LLMs, Transformers, vision-language models, fine-tuning, calibration and abstention, quantum machine learning

Frameworks and Libraries: PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, Transformers, OpenCV, CLIP, MediaPipe, PennyLane, lambeq, Hydra

Software and Web: Django, React, Node.js, JavaFX, REST APIs, WebSocket, Docker, CI/CD, OOP, design patterns, Agile/Scrum, unit testing

Databases: MySQL, PostgreSQL, SQLite, MongoDB

Embedded and Graphics: STM32, embedded C, ADC-DMA acquisition, Godot 4, Blender, SDL3, CMake

Tools: Git, Linux, VS Code, Jupyter, LaTeX, Ollama, Apple MPS

Extracurricular Activities

IEEE NSU Student Branch, General Member **2023 – Present**

Technical workshops, seminars and community outreach programmes.

References

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